

DSC540Week9&10

May 19, 2025

1 Activity For Week 9 & 10

1.0.1 Activity 7.01

```
[2]: import urllib.request, urllib.parse, urllib.error
import requests
from bs4 import BeautifulSoup
import ssl
import re
```

```
[4]: # Read the HTML from the URL and pass on to BeautifulSoup
top100url = 'https://www.gutenberg.org/browse/scores/top'
response = requests.get(top100url)
```

```
[6]: # Function to check the status for the web URL
def status_check(r):
    if r.status_code==200:
        print("Success!")
        return 1
    else:
        print("Failed!")
        return -1
```

```
[8]: status_check(response)
```

Success!

```
[8]: 1
```

```
[10]: # Reading the content from html page

contents = response.content.decode(response.encoding)
soup = BeautifulSoup(contents, 'html.parser')
```

```
[12]: # Empty list to hold all the http links in the HTML page

lst_links=[]

# Find all the href tags and store them in the list of links
```

```

for link in soup.find_all('a'):
    #print(link.get('href'))
    lst_links.append(link.get('href'))

```

```
[14]: lst_links[:30]
```

```

[14]: ['/',
      '/about/',
      '/about/',
      '/policy/collection_development.html',
      '/about/contact_information.html',
      '/about/background/',
      '/policy/permission.html',
      '/policy/privacy_policy.html',
      '/policy/terms_of_use.html',
      '/ebooks/',
      '/ebooks/',
      '/ebooks/categories',
      '/ebooks/bookshelf/',
      '/browse/scores/top',
      '/ebooks/offline_catalogs.html',
      '/help/',
      '/help/',
      '/help/copyright.html',
      '/help/errata.html',
      '/help/file_formats.html',
      '/help/faq.html',
      '/policy/',
      '/help/public_domain_ebook_submission.html',
      '/help/submitting_your_own_work.html',
      '/help/mobile.html',
      '/attic/',
      '/donate/',
      '/donate/',
      'pretty-pictures',
      '#books-last1']

```

```
[22]: booknum=[]
```

```
[24]: # looping list to find the book numbers only from text.
```

```

for i in range(19,119):
    link=lst_links[i]
    link=link.strip()
    # Regular expression to find the numeric digits in the link (href) string
    n=re.findall('[0-9]+',link)

```

```

if len(n)==1:
    # Append the filename casted as integer
    booknum.append(int(n[0]))

```

```

[26]: print ("\nThe file numbers for the top 100 ebooks on Gutenberg are shown_
        ↳below\n"+"-"*70)
print(booknum)

```

The file numbers for the top 100 ebooks on Gutenberg are shown below

```

-----
[1, 1, 7, 7, 30, 30, 1023, 84, 2701, 1342, 1513, 11, 2542, 64317, 844, 174, 100,
145, 37106, 2641, 43, 2554, 16389, 67979, 345, 98, 26184, 394, 6761, 6593, 4085,
2160, 1260, 1259, 5197, 19623, 19616, 19622, 1080, 19619, 19624, 19620, 19618,
19617, 19621, 1998, 5200, 3296, 1400, 16328, 19625, 76, 25344, 1661, 28054,
20029, 1232, 3207, 2591, 41445, 1952, 4300, 45, 4363, 6130, 408, 2000, 1184, 74,
19626, 2600, 219, 768, 34901, 76072, 205, 1727, 36034, 135, 5740, 1497, 7370,
8800, 244, 35899, 120, 30254, 76076, 76075, 76071]

```

```

[28]: print(soup.text[:2000])

```

Top 100 | Project Gutenberg

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Top 100 EBooks yesterday -
Top 100 Authors yesterday -

Top 100 EBooks last 7 days -
Top 100 Authors last 7 days -
Top 100 EBooks last 30 days -
Top 100 Authors last 30 days

Top 100 EBooks yesterday

Bleak House by Charles Dickens (14635)
Frankenstein; Or, The Modern Prometheus by Mary Wollstonecraft Shelley (5191)
Moby Dick; Or, The Whale by Herman Melville (3330)
Pride and Prejudice by Jane Austen (2118)
Romeo and Juliet by William Shakespeare (1914)
Alice's Adventures in Wonderland by Lewis Carroll (1855)
A Doll's House : a play by Henrik Ibsen (1648)
The Great Gatsby by F. Scott Fitzgerald (1563)
The Importance of Being Earnest: A Trivial Comedy for Serious People by Oscar Wilde (1506)
The Picture of Dorian Gray by Oscar Wilde (1477)
The Complete Works of William Shakespeare by William S

```
[30]: # Temp empty list of Ebook names  
lst_titles_temp=[]
```

```
[32]: # Getting start index from website.  
  
start_idx=soup.text.splitlines().index('Top 100 EBooks yesterday')
```

```
[34]: # Looping through to get the title details.  
  
for i in range(100):  
    lst_titles_temp.append(soup.text.splitlines()[start_idx+2+i])
```

```
[36]: lst_titles=[]  
for i in range(100):  
    id1,id2=re.match('[a-zA-Z ]*',lst_titles_temp[i]).span()  
    lst_titles.append(lst_titles_temp[i][id1:id2])
```

```
[38]: for l in lst_titles:  
        print(l)
```

Bleak House by Charles Dickens
Frankenstein
Moby Dick
Pride and Prejudice by Jane Austen
Romeo and Juliet by William Shakespeare
Alice
A Doll
The Great Gatsby by F

The Importance of Being Earnest
The Picture of Dorian Gray by Oscar Wilde
The Complete Works of William Shakespeare by William Shakespeare
Middlemarch by George Eliot
Little Women
A Room with a View by E
The Strange Case of Dr
Crime and Punishment by Fyodor Dostoyevsky
The Enchanted April by Elizabeth Von Arnim
The Blue Castle
Dracula by Bram Stoker
A Tale of Two Cities by Charles Dickens
Simple Sabotage Field Manual by United States
Cranford by Elizabeth Cleghorn Gaskell
The Adventures of Ferdinand Count Fathom
History of Tom Jones
The Adventures of Roderick Random by T
The Expedition of Humphry Clinker by T
Jane Eyre
Twenty years after by Alexandre Dumas and Auguste Maquet
My Life
Aesop
Aesop
Aesop
A Modest Proposal by Jonathan Swift
Aesop
Aesop
Aesop
Aesop
Aesop
Aesop
Aesop
Thus Spake Zarathustra
Metamorphosis by Franz Kafka
The Confessions of St
Great Expectations by Charles Dickens
Beowulf
Aesop
Adventures of Huckleberry Finn by Mark Twain
The Scarlet Letter by Nathaniel Hawthorne
The Adventures of Sherlock Holmes by Arthur Conan Doyle
The Brothers Karamazov by Fyodor Dostoyevsky
Las F
The Prince by Niccol
Leviathan by Thomas Hobbes
Grimms
Frankenstein
The Yellow Wallpaper by Charlotte Perkins Gilman
Ulysses by James Joyce

Anne of Green Gables by L
Beyond Good and Evil by Friedrich Wilhelm Nietzsche
The Iliad by Homer
The Souls of Black Folk by W
Don Quijote by Miguel de Cervantes Saavedra
The Count of Monte Cristo by Alexandre Dumas and Auguste Maquet
The Adventures of Tom Sawyer
Aesop
War and Peace by graf Leo Tolstoy
Heart of Darkness by Joseph Conrad
Wuthering Heights by Emily Bront
On Liberty by John Stuart Mill
The mystery of Angelina Frood by R
Walden
The Odyssey by Homer
White Nights and Other Stories by Fyodor Dostoyevsky
Les Mis
Tractatus Logico
The Republic by Plato
Second Treatise of Government by John Locke
The divine comedy by Dante Alighieri
A Study in Scarlet by Arthur Conan Doyle
The Philippines a Century Hence by Jos
Treasure Island by Robert Louis Stevenson
The Romance of Lust
Affirmations by Havelock Ellis
The Little Review
Bed
Moby Word Lists by Grady Ward
The Works of Edgar Allan Poe
Cookery and Dining in Imperial Rome by Apicius
The Wonderful Wizard of Oz by L
Du c
The King in Yellow by Robert W
Oliver Twist by Charles Dickens
Anna Karenina by graf Leo Tolstoy
Little Women by Louisa May Alcott
A child
A Christmas Carol in Prose
The Reign of Greed by Jos
A loyal lass
Gulliver
Dubliners by James Joyce
The Legend of Sleepy Hollow by Washington Irving

1.1 Activity 7.02

```
[42]: import urllib.request, urllib.parse, urllib.error
import json
```

1.2 Load the secret API key (you have to get one from the OMDb website and use that; it has a 1,000 daily limit) from a JSON file, stored in the same folder into a variable, by using `json.loads()`.

Note The following cell will not be executed in the solution notebook because the author cannot give out their private API key. The students/users/instructor will need to obtain a key and store it in a JSON file. We are calling this file `APIkeys.json`.

```
[95]: # Got the API Key to read the data using OMDbApi.
with open('APIkeys.json') as f:
    keys = json.load(f)
    omdbapi = keys['OMDBApi']
```

```
[97]: serviceurl = 'http://www.omdbapi.com/?'
```

```
[99]: apikey = '&apikey='+omdbapi
```

```
[102]: # function to save the poster details.
def save_poster(json_data):
    import os
    title = json_data['Title']
    poster_url = json_data['Poster']
    # Splits the poster url by '.' and picks up the last string as file_
    ↪extension
    poster_file_extension=poster_url.split('.')[ -1]
    # Reads the image file from web
    poster_data = urllib.request.urlopen(poster_url).read()

    savelocation=os.getcwd()+ '\\'+ 'Posters'+ '\\ '
    # Creates new directory if the directory does not exist. Otherwise, just_
    ↪use the existing path.
    if not os.path.isdir(savelocation):
        os.mkdir(savelocation)

    filename=savelocation+str(title)+'.'+poster_file_extension
    f=open(filename, 'wb')
    f.write(poster_data)
    f.close()
```

```
[104]: # function to search movies in OMDb api
def search_movie(title):
    try:
        url = serviceurl + urllib.parse.urlencode({'t': str(title)})+apikey
        print(f'Retrieving the data of "{title}" now... ')
    except:
```

```

print(url)
uh = urllib.request.urlopen(url)
data = uh.read()
json_data=json.loads(data)

if json_data['Response']=='True':
    print_json(json_data)
    # Asks user whether to download the poster of the movie
    if json_data['Poster']!='N/A':
        save_poster(json_data)
else:
    print("Error encountered: ",json_data['Error'])

except urllib.error.URLError as e:
    print(f"ERROR: {e.reason}")

```

[106]: `search_movie("Titanic")`

Retrieving the data of "Titanic" now...

<http://www.omdbapi.com/?t=Titanic&apikey=d5b9fb85>

```

-----
Title: Titanic
Year: 1997
Rated: PG-13
Released: 19 Dec 1997
Runtime: 194 min
Genre: Drama, Romance
Director: James Cameron
Writer: James Cameron
Actors: Leonardo DiCaprio, Kate Winslet, Billy Zane
Plot: A seventeen-year-old aristocrat falls in love with a kind but poor artist
aboard the luxurious, ill-fated R.M.S. Titanic.
Language: English, Swedish, Italian, French
Country: United States
Awards: Won 11 Oscars. 126 wins & 83 nominations total
Ratings: [{'Source': 'Internet Movie Database', 'Value': '7.9/10'}, {'Source':
'Rotten Tomatoes', 'Value': '88%'}, {'Source': 'Metacritic', 'Value': '75/100'}]
Metascore: 75
imdbRating: 7.9
imdbVotes: 1,334,872
imdbID: tt0120338
-----

```

[108]: `search_movie("Random_error")`

Retrieving the data of "Random_error" now...

http://www.omdbapi.com/?t=Random_error&apikey=d5b9fb85

Error encountered: Movie not found!

1.3 2. Activity for Week 7 & 8 Dataset for Visualization

```
[126]: import pandas as pd
import numpy as np
import seaborn as sns
from fuzzywuzzy import process, fuzz
import matplotlib.pyplot as plt
import matplotlib.pyplot as pl
sns.set_style("whitegrid")
sns.set_palette("RdBu")
```

```
[128]: # read candy hierarchy data set

candyData = pd.read_excel("./candyhierarchy2017.xlsx")
```

```
/opt/anaconda3/lib/python3.12/site-packages/openpyxl/worksheet/_read_only.py:81:
UserWarning: Unknown extension is not supported and will be removed
    for idx, row in parser.parse():
```

```
[132]: # handle columns name

# 1- create dictionary key is old name and value is new name
newNames = {"Q1: GOING OUT?" : "going out", "Q2: GENDER": "gender", "Q3: AGE": "age",
            "Q4: COUNTRY": "country",
            "Q10: DRESS" : "dress", "Q12: MEDIA [Science]": "Science", "Q11: DAY": "day",
            "Q12: MEDIA [Daily Dish]": "Daily Dish",
            "Q12: MEDIA [ESPN]": "ESPN", "Q12: MEDIA [Yahoo]": "Yahoo"}

# 2- rename columns
candyData.rename(columns = newNames, inplace= True)
```

```
[134]: # show unique values

candyData["going out"].unique()
```

```
[134]: array([nan, 'No', 'Yes'], dtype=object)
```

```
[136]: # create function that visualized categorical column using pie plot

def pie_plot(column_name, explodeIndex = None):
    """
    1) input : column name, column data type must be object or categorical
    2) explodeIndex, is the index i need to explode it
    2) output : circle chart that shows size of each unique values and
    percentage
    """
    # Create explode list with zeros of size equal to the number of unique
    values
```

```

explodeList = [0] * candyData[column_name].nunique()

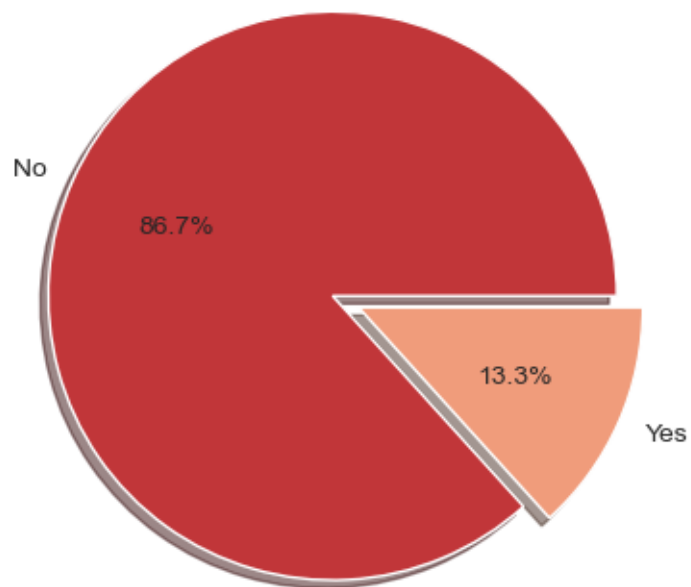
# Check and set explodeIndex value
if explodeIndex is not None:
    explodeList[explodeIndex] = 0.1

# Create pie plot
plt.pie(candyData[column_name].value_counts(), labels = _
candyData[column_name].value_counts().index, shadow = True, autopct = "%1.
1f%%", explode = explodeList)
plt.show()

```

[138]: # visualized going out column using pie plot function i create it in cell 15

```
pie_plot(column_name = "going out", explodeIndex = 0)
```



[142]: # Survery respondents intending to 'GO OUT' Trick-or-Treating for Halloween.

```

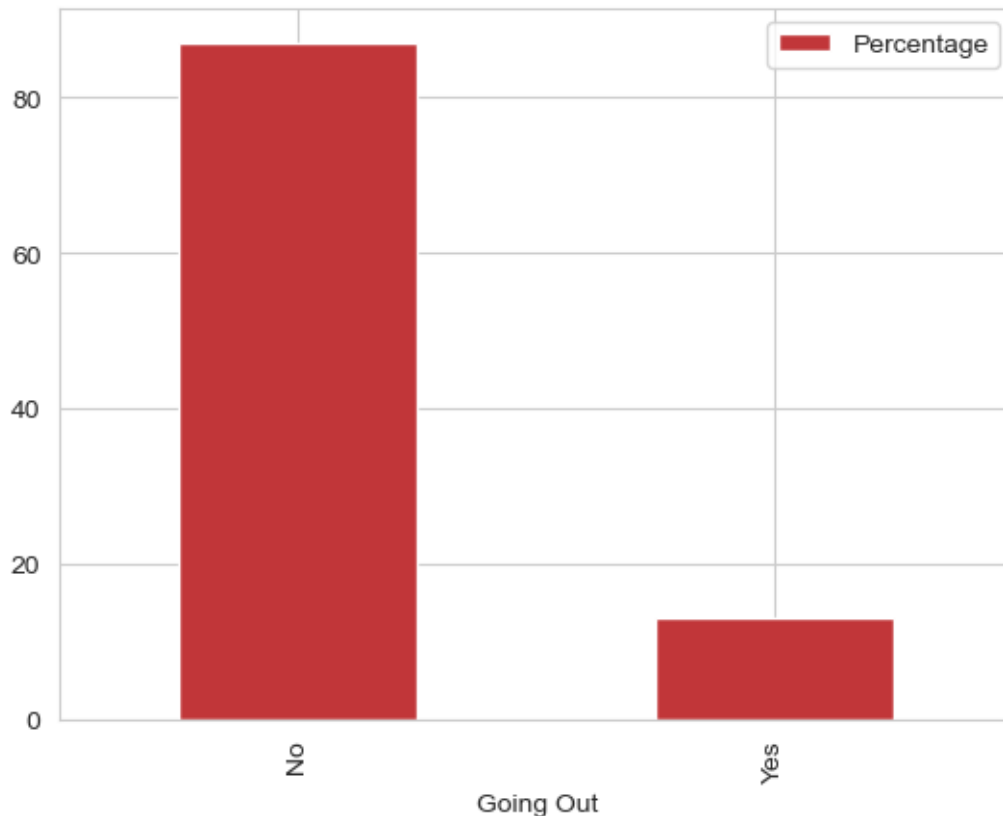
i= {'Going Out': ['No','Yes'], 'Count': [2008,298]}
I= pd.DataFrame.from_dict(i)
I['Percentage']= round((I['Count'] / I['Count'].sum()) * 100,2)

```

```
[144]: # let's bar chart it to see how it looks like in there
```

```
I.plot(kind='bar', x='Going Out', y='Percentage')
```

```
[144]: <Axes: xlabel='Going Out'>
```



```
[146]: # see number of missing values
```

```
candyData["gender"].isna().sum()
```

```
[146]: 41
```

```
[148]: # fill missing values
```

```
candyData["gender"].fillna("Other", inplace = True)
```

```
/var/folders/18/ygzmjv90m72g8_1d1sfk1b80000gn/T/ipykernel_2516/2581853504.py:3:
FutureWarning: A value is trying to be set on a copy of a DataFrame or Series
through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work
because the intermediate object on which we are setting values always behaves as
```

a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.

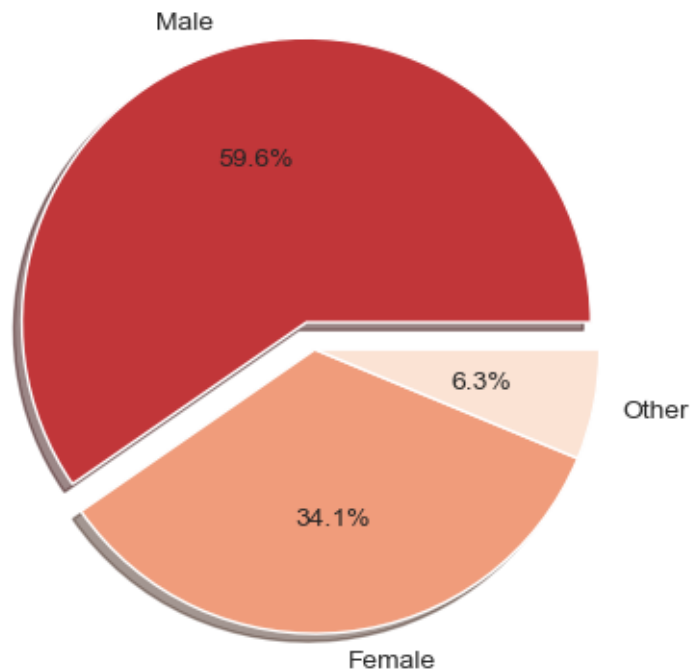
```
candyData["gender"].fillna("Other", inplace = True)
```

```
[150]: # replace Other with I'd rather not say
```

```
candyData["gender"].replace("I'd rather not say", "Other", inplace = True)
```

```
[152]: # visualized gender column using pie plot function i create it in cell 15
```

```
pie_plot(column_name = "gender", explodeIndex = 0)
```



1.4 3. Connect to an API for Brewery company and pulling all the required details in dataframe in table format.

```
[21]: import pandas as pd  
import requests
```

```
[86]: import requests
import json

url = "https://api.openbrewerydb.org/v1/breweries?by_city=miami" # Example URL

response = requests.get(url)

if response.status_code == 200: # Check for a successful response
    json_data = response.json() # Parse the JSON response
    df = pd.json_normalize(json_data)
    df = df.drop('id',axis=1)
    # Print the DataFrame

    print(df)

else:
    print("Error:", response.status_code)
```

	name	brewery_type	address_1 \
0	Abbey Brewing Co	contract	1115 16th St
1	Bay Brewing Company	planning	None
2	Biscayne Bay Brewing Co	micro	8000 NW 25th St Ste 500
3	Bob Loblaw Brewing	planning	None
4	Bousa Brewing Company	micro	7235 NE 4th Ave
5	Cervecentro	planning	None
6	Cerveza Guajira	planning	None
7	Concrete Beach Brewery	micro	325 NW 24th St
8	D. MANATEE HOLDINGS, LLC	planning	None
9	Gravity Brewlab	contract	1521 Alton Rd # 591
10	J Wakefield Brewing	micro	120 NW 24th St
11	Lincoln's Beard Brewing Co.	micro	7360 SW 41st St
12	Lucky Star Brewery	brewpub	219 S 2nd St
13	NightLife Brewing Co	micro	1588 Nw 7th St
14	Star City Brewing Company	micro	319 S 2nd St
15	Sweet Mercy Brewing Co.	planning	None
16	The Tank Brewing Company	micro	5100 NW 72nd Ave. Bay A-1
17	Titanic Brewing Co	brewpub	5813 Ponce de Leon Blvd
18	Veza Sur Brewing Co	large	55 NW 25th St
19	Wynwood Brewing Company	micro	565 NW 24th St

	address_2	address_3	city	state_province	postal_code	country \
0	None	None	Miami Beach	Florida	33139-2441	United States
1	None	None	Miami	Florida	33130-3488	United States
2	None	None	Miami	Florida	33122-1631	United States
3	None	None	Miami	Florida	33179	United States
4	None	None	Miami	Florida	33138-5315	United States
5	None	None	Miami	Florida	33181-3404	United States
6	None	None	Miami	Florida	33137	United States

7	None	None	Miami	Florida	33127-4325	United States
8	None	None	Miami	Florida	33131-1210	United States
9	None	None	Miami Beach	Florida	33139-3301	United States
10	None	None	Miami	Florida	33127-4414	United States
11	None	None	Miami	Florida	33155-4504	United States
12	None	None	Miamisburg	Ohio	45342-2924	United States
13	None	None	Miami	Florida	33125	United States
14	None	None	Miamisburg	Ohio	45342-2926	United States
15	None	None	Miami	Florida	33143	United States
16	None	None	Miami	Florida	33166	United States
17	None	None	Miami	Florida	33146-2422	United States
18	None	None	Miami	Florida	33127-4415	United States
19	None	None	Miami	Florida	33127-4327	United States

	longitude	latitude	phone	website_url \
0	-80.140294	25.789038	3055388110	None
1	NaN	NaN	18134763767	None
2	NaN	NaN	None	None
3	NaN	NaN	3052495628	http://www.bobloblawbrewing.com
4	-80.189410	25.841459	3053635166	http://www.bousabrewing.com
5	NaN	NaN	3055171656	http://www.cervecentro.com.ve
6	NaN	NaN	3056005692	None
7	-80.200917	25.800254	3057962727	http://www.concretebeachbrewery.com/
8	NaN	NaN	None	http://www.tiliaventure.com
9	NaN	NaN	7865367085	http://www.Gravity.beer
10	-80.197529	25.799544	7862547779	http://www.jwakefieldbrewing.com
11	-80.314353	25.732096	3059127390	http://lincolnsbeardbrewing.com/
12	-84.287824	39.639436	9378662739	http://www.luckystarbrewery.com
13	-80.221313	25.780069	7867872337	http://www.nightlifebrewingco.com
14	-84.288596	39.637994	9377017827	http://www.starcitybrewing.com
15	NaN	NaN	3057999260	http://www.sweetmercybrewing.com
16	NaN	NaN	7868011554	http://www.thetankbrewing.com
17	-80.271314	25.719812	3056672537	http://www.titanicbrewery.com
18	-80.196149	25.800897	None	None
19	-80.204493	25.800376	3056405043	http://www.wynwoodbrewing.com

	state	street
0	Florida	1115 16th St
1	Florida	None
2	Florida	8000 NW 25th St Ste 500
3	Florida	None
4	Florida	7235 NE 4th Ave
5	Florida	None
6	Florida	None
7	Florida	325 NW 24th St
8	Florida	None
9	Florida	1521 Alton Rd # 591
10	Florida	120 NW 24th St

11	Florida	7360 SW 41st St
12	Ohio	219 S 2nd St
13	Florida	1588 Nw 7th St
14	Ohio	319 S 2nd St
15	Florida	None
16	Florida	5100 NW 72nd Ave. Bay A-1
17	Florida	5813 Ponce de Leon Blvd
18	Florida	55 NW 25th St
19	Florida	565 NW 24th St

[]: